Manual

Process Data Management

PDV-PDM 8.2

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Contents

1 Overview: Process Data Management

Overview

Purpose

This function package includes the user interface to define master data for Process Data Collection. The function package also contains the service for storing collected measured values and for generating samples for statistical process data reports. Process Data Collection can be set up to collect TAG-based entry processes for specific machines. This, among other things, is a requirement for ID tracing (PDV-PTR).

The PDV-VRP function package "Processing rules for process data" is required to control measured values processed by applications.

Integration

This function package requires a machine interface for data collection in the HYDRA Process Communication Controller (PCC).

Features

Extensive configuration functions to create and edit master data for process values

- Configuration of logical channels to collect process values
- Definition of input and output channels
- Extensive functions to create and edit machine and article-related monitoring parameters for process values (lower/ upper action limits, lower/ upper tolerance limits ...)
- Functions for setting parameters for statistical sampling functions and sample generation
- List to manage specific monitoring parameters depending on certain machines, articles and tools
- List defining binary process events
- Sample generation at specified sampling intervals. Intervals are specified when defining collection rules for characteristics
- Based on these rules, the service generates sample data available for advanced sample reports

- Collection of text tags to identify measured values. Optimized storage of related measured values. The machines specify their relation and save it in the identification tags irrespective of the collection time.
- Service indexing measured values according to the defined identification tags

2 Logical channels

Overview

Menu	Master data → Process data processing → Logical channels
Transaction code	lgchcnf
Function authorization	lgchcnf.*

The logical channels form the central point where logical configurations, physical machines and shop floor servers (terminal type PCC) come together.

Purpose

You must configure the logical channels in order to assign the process parameters/PDV events:
- to machines
- to shop floor servers (PCC)
- to physical channel numbers.
The logical channels define which technical channel is used on a terminal to record the characteristics. Logical channels specify the "channel mapping".

You can also use this configuration to specify the physical channels in more detail, for example:
- you can specify how data is collected
- you can define outputs e.g. for alerts or
- you can define target values.
As of PDV 8.2, you can also configure the channels for the recording of TAGs (e.g. serial number, etc.). This way, you can connect the measured values to an alphanumeric TAG and you can perform analyses based on TAGs in defined evaluations.

Logical channels are versioned master data. For each data record, a validity period is therefore specified (exact to the second). Only in the period specified, the data record is valid for the system.

Note: A modification of the channel configuration is not passed on to the terminal, i.e. the new channel configuration is made available to the terminal only after the terminal/ shop floor components are restarted or after the cyclical start of the configuration monitor service (e.g. in certain situations such as target value modifications).

Integration

The logical channels are used to assign process parameters or PDV events to a machine-shop floor server combination.

Selection criteria

The application provides the following selection criteria:

"General" tab

Channel

Selection of channel number

Designation

Selection of channel designation including wildcard function and search screen.

Channel type

Selection of channel type using selection box

Data class of the channel

Selection of data class using selection box

"Assignment" tab

Machine

Selection of machines including wildcard function and search screen

PCC terminal

A selection of terminal/shop floor server is possible. A search screen may also be used.

"Activation/alert" tab

Active

Selection of active channels only

Alert

Selection of logical channels with enabled option *Alert*

Valid from-to

Start and end time including date and time to specify the validity period

Alert channel

Selection of the logical channels where the respective alert channel is stored.

Field descriptions

Tab "General"

Channel

Channel number on the machine. This number must be unique for the PCC terminals. Only values between 1 and 9999 are permitted.

Designation

Optional, logical designation of the logical channel

Channel type

Includes the reference to the data that is collected via the channel. This can either be a configured PDV event (`E`) or a process parameter (`PP`, corresponds to the decimal value or the alphanumeric tag) of a characteristic.

Data class of the channel

For channels of type `PP`, the data class of the channel defines which value of a process parameter is collected using the channel. The values supported are:

- MV: Measured value/ alphanumeric tag
- TV: Target value
- UTL: Upper tolerance limit
- UPAL: Upper process action limit
- LPAL: Lower process action limit
- LTL: Lower tolerance limit

If a target value or a tolerance or process action limit is passed via a channel, it must be guaranteed that a time span of at least 30 minutes is respected between two value changes of a channel.

Data type (available as of PDV 8.2)

Data type that is recorded via the channel. The following three options are available:

- Decimal
- Alphanumeric
- Tag (alphanumeric)

The data types *decimal* and *alphanumeric* are used for the typical process parameters. The tag values recorded are available as selection criteria in the selection criteria based on tags. These values can be selected.

The permitted characters for the data types are defined as follows:

- Decimal: Numeric decimal values
- Tag (alphanumeric): 0-9 a-z A-Z äöüÄÖÜ _-+#;.,

Note:

The recording of tags is limited to 50 characters. If the data source transfers a tag with more than 50 characters (alphanumeric), the shop floor client cuts off the values after 50 characters.

Tab "Assignment"

Machine

Assignment of the machine

PCC terminal

Assignment of the PCC terminal

Process parameter

Assignment of a process parameter if the channel type is defined for the collection of a process parameter.

Event

Assignment of a defined event if the channel type is defined for the collection of an event.

Tab "Properties"

Input type

The input type of the channel is used to define how the shop floor server is to access the machine data. The control parameters are used for the collection of process values and of events (see parameter type). The values supported are:

- A: Automatic (if supported by driver)
- T: Trigger controlled
- C: Cyclic

Cycle time

If the channel is controlled using a cycle time, this field contains the number of seconds after which the values are queried.

Trigger

If the channel is controlled via trigger channel, the channel number of the trigger is stored in this field. Only values between 1 and 9999 are permitted.

Direction

Direction of data flow for the channel. Values can be transmitted from the machine to the data collection (input, column value `I`). This is the direction normally used to record machine data. In addition, the data direction can be configured from the data collection to the machine (output, column value `O`) to overwrite a value depending on the selected data class of the channel. For example, setting a new target value for the connected machine.

Tab "Activation/alert"

Active

The logical channel is switched to the active status here. In data collection, only active and valid channels are used. The activation or deactivation of a channel is only valid after the restart of the respective shop floor components.

Valid from-to

Validity period during which the channel is used for data collection.

Alert

This option specifies if a configured alert channel is activated (output signal for setting a physical signal). The activation or deactivation of a channel is only valid after the restart of the respective shop floor components.

Alert channel

This option specifies the physical channel number where an output signal is set when an alert condition is fulfilled. Depending on the channel type, the alert can be triggered by a PDV event or a limit violation of a process parameter. Only values between 1 and 999 are permitted.

Configuration of an alert channel

In order to activate an output signal in case of a limit value violation of a process parameter, an output channel of the type "Process parameter" must be created.

The following settings are made in the dialog "Insert logical channel".

- For the channel number, you configure a "virtual channel' with any value (between 1 and 9999). However, this number must be unique. This channel number is used to manage the combination "Alert channel – data class of the channel".
- You use the "Data class of the channel" to define the limit value violation that triggers the alert.
- You must also specify the process parameter that is monitored.
- Activate "Output" as direction.
- Enable the options "Channel active" and "Alert active".
- The alert channel must match the physical channel where an output signal is set in case of an alert condition.

For a detailed description of individual input fields, refer to section "Field description".

Checking Business Parameter Containers (BSCs)

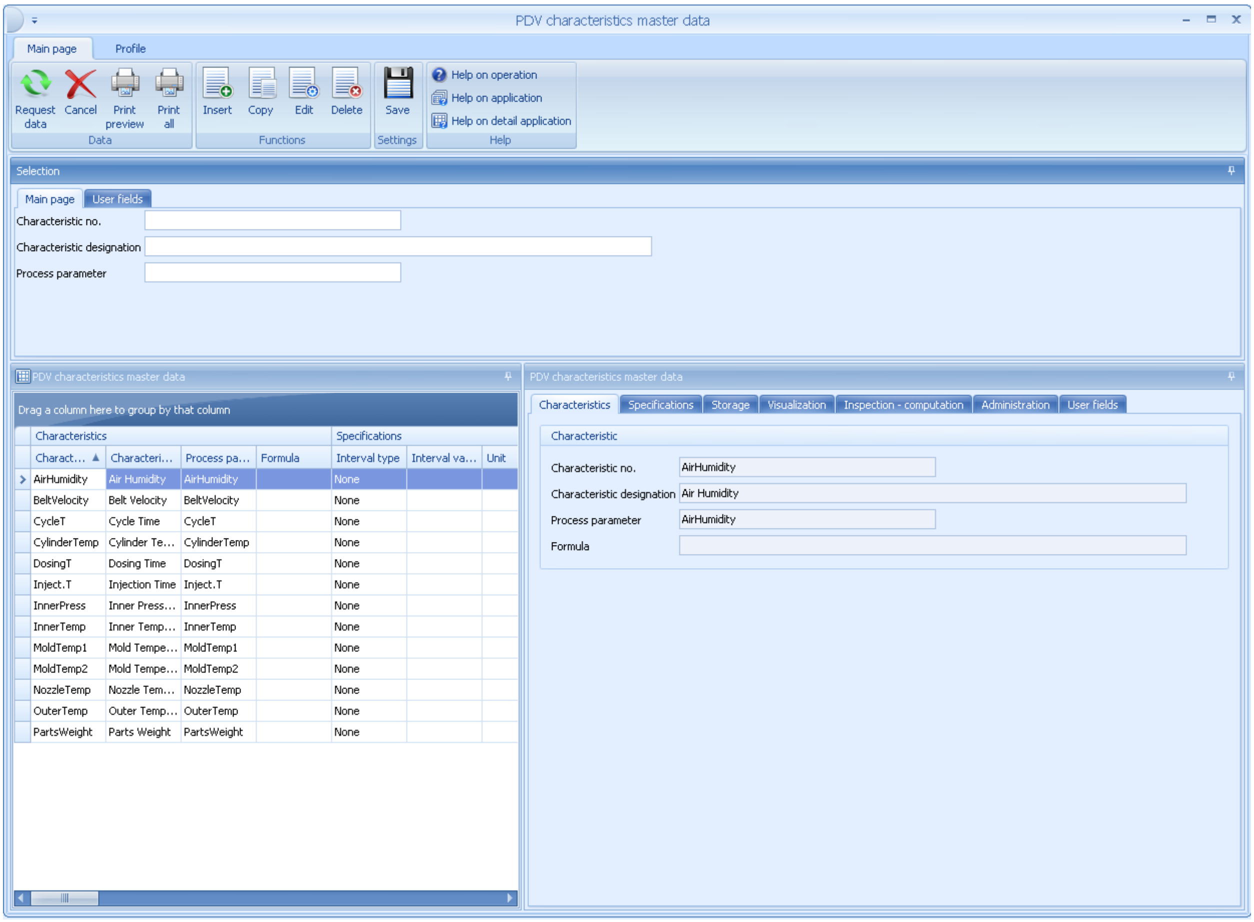
See here for further information on checking the system with respect to business parameters.

3 PDV Characteristics Master Data

Overview

Menu	Master data → Process data processing → PDV characteristics master data
Transaction code	chrp
Function authorization	chrq.*

You use the catalog of characteristics to predefine characteristics that are then used in collection rules. It aims at persons involved in data collection planning. The characteristics catalog is one of the most important basic catalogs. You need the characteristics catalog to create collection rules. As this catalog is used to predefine characteristics data for collection rules, it includes extensive input options.



Purpose

The characteristics catalog is one of the most important basic catalogs. You need the characteristics catalog to create collection rules. As this catalog is used to predefine characteristics data for collection rules, it includes extensive input options. Only enter data in the characteristics catalog that need not be changed when later assigned to a collection rule. Do not define limit values, for example. This is normally not useful because the limit values are only known when the collection rule is created. Only when you create and assign a collection rule, a relation to a concrete article or machine is established. Note this and you will know what kind of information you should predefine. For example, it must be carefully considered whether the characteristic "outer diameter" is only created once and detailed information is stored in the collection rule later on or whether several "outer diameter characteristics" are created, e. g. with specification of limit values. Usually, it is an advantage to store a small number of general characteristics. The required evaluations/reports also play a role in this context. If a new "outer diameter characteristic" is created for almost every tolerance change, this characteristic is "valid" for one article or machine only. In a subsequent failure analysis, a comprehensive evaluation is not possible in this case!

It is important that you can still change any definition stored in the catalog or add missing definitions when you later plan the collection rule.

Important

The configurations made in the characteristics' master data need not be definitive for the collection rule. The characteristics' master data is used as a template when the collection rules are later created. You can complete and change all configurations of the characteristics' master data in the collection rule.

Integration

The characteristics' master data is used as a template when the collection rules are later created. The field *Process parameter* provides a logical connection to the logical channels. The connection is only established when the characteristics are definitively defined in the collection rules.

Selection criteria

The application provides the following selection criteria:

- Characteristic no.:
  Characteristic number

- Designation:
  Name of the characteristic

- Process parameter:
  Process parameter

If several selection criteria are used, the overlapping results are displayed below in the *Characteristics master data.*

Field descriptions

Find below a description of the columns and input options for characteristics:

Tab *Characteristics*

Characteristic no.

Unique number of the characteristic

Characteristic designation

Name of the characteristic

Process parameter

Predefinition of the process parameter

Formula

See section "Formula calculation" below.

Tab *Specifications*

Select the *Specifications* tab to enter the sampling scheme and the tolerance limits. Note: As mentioned above, it is only reasonable to define tolerance limits in the characteristics master data under certain circumstances.

Sampling scheme (no longer relevant as of PDV 8.3)

The following sampling schemes are available:

- None
- Piece-related

With a piece-related sampling scheme, you can define the interval. All single values collected in this interval are then combined and form one sample. If you define a piece-related sampling scheme, you can make evaluations for samples, for example control charts.

Interval value (no longer relevant as of PDV 8.3)

Interval used to combine the single values collected in this interval to a sample.

Display of failures in the Failure Mode Analysis (PDV 8.1 and PDV 8.2)

To display the created failures in the Failure Mode Analysis, you must specify an interval value. Process parameters without specified interval value are not integrated in the Failure Mode Analysis.

Field visibility

The field is only visible and input is only possible if the user is assigned the function authorization "InspectionInterval".

Unit

Pieces, meter, kg, etc. Units are assigned using the unit catalog.

Decimal places (no longer relevant as of PDV 8.1)

Enter the decimal places. Leading zeros before the comma are not displayed in the specification fields. By default, the number of decimal places defined in the system settings is pre-assigned.

Size (measure type)

Validation and tolerance limits can be entered as absolute, relative or percentage values. Note: You must enter relative or percentage limits as negative values.

Upper PL

Specifies the upper process limit. In PDV, this value also defines the displayed upper red area in process visualization.

Upper TL

Specifies the upper tolerance limit (upper specification limit)

Target value

Specifies the target value

Lower TL

Specifies the lower tolerance limit (lower specification limit)

Lower PL

Specifies the lower process limit. In PDV, this value also defines the displayed lower red area in process visualization.

Upper TL – Generate failure / Lower TL

If the checkboxes *Generate failure* are enabled, a violation of the limit value automatically results (in the background) in the generation of a failure with failure type "limit value violation" (AUTO:TG> or AUTO:TG<) when measured values are collected. The generated failures are evaluated in the Failure Mode Analysis, for example.

Specifications process

Upper PAL

Upper process action limit

Lower PAL

Lower process action limit

Generate failure

If the checkboxes *Generate failure* are enabled, a violation of the limit value automatically results (in the background) in the generation of a failure with failure type "limit value violation" (AUTO:PEG> or AUTO:PEG<) when measured values are collected. The generated failures are evaluated in the Failure Mode Analysis, for example.

Storage (no longer relevant as of PDV 8.3)

Using the filter functions for storage, you define the frequency used to store single values. This value affects the further processing of measured values on the HYDRA server. In later evaluations, only the actually stored values can be displayed. The storage frequency is also the basis for further aggregations by the PDV Distributor calculating samples. Here, the PDV Distributor only accesses data of the online data set.

Example: The defined filter specifies that only one of ten measured values is saved. At the same time, a sample interval specifies that 5 measured values are combined to one sample. After filtering, the storage frequency is the basis for the calculation of samples. 5 measured values are then actually combined to one sample. Result: For 50 measured values, which have actually been collected, one sample is generated.

Filter function:

Define a filter function for the collection of PDV values. The following four filter functions are available: `None`, `Cyclic`, `Frequency` and `Percentage`. Depending on the selection made, different input fields are shown below the combo box. Use these fields to parameterize the filter function.

- `None`
  No filter, the system saves each collected value.
- `Frequency`
  Enter a number of measured values. This means that the system only saves every n measured value. The system evaluates the measured values collected in the meantime but they are not saved.
- `Cyclic`
  Enter a time interval in seconds. This means that the system only saves a measured value every n seconds. The system evaluates the measured values collected in the meantime but they are not saved.
- `Percentage`
  Enter a time interval and a percentage. In the PDV data collection, the system saves the

measured value every n seconds or if the absolute deviation of the current measurement compared to the previous measurement exceeds the percentage entered here. The time interval is reset when the measured value is saved because of the percentage deviation. The system evaluates the measured values collected in the meantime but they are not saved.

Visualization

Visualization

Check this option to enable the online visualization for the process parameter.

Position

Use this integer field to specify the visualization position of the process parameter. Use this for graphics in the online visualization with more than one display element on one page. Example: If 16 display elements are shown on one page, you can use this option to define the position of the different process parameters.

If the value 0 or less is specified, the process parameter is not visualized. You can only edit the input field after you have checked the option *Visualize*.

Visualization *Filter function*

Here, the same filter functions are available as for the storage, but you can use this second setting to decouple the storage from the online visualization. For example, you can configure that each measured value is stored, but only every tenth measured value is visualized.

The following four filter functions are available: `None`, `Cyclic`, `Frequency` and `Percentage`. Depending on the selection made, different input fields are shown below the combo box. Use these fields to parameterize the filter function.

- `None`

  No filter, the system visualizes each collected value.

- `Frequency`

  Enter a number of measured values. This means that the system only visualizes every n measured value.

- `Cyclic`

  Enter a time interval in seconds. This means that the system only visualizes a measured value every n seconds.

- `Percentage`

  Enter a time interval and a percentage. In the PDV data collection, the system visualizes a measured value every n seconds or if the absolute deviation of the current measurement compared to the previous measurement exceeds the percentage entered here. The time interval is reset when the measured value is visualized because of the percentage deviation.

Inspection – computation

Check characteristic (no longer relevant as of PDV 8.3)

You can use this option to define whether the process parameter is checked against limit value violations in the PDV data collection or whether the collected measured values only pass the data collection to be stored.

This field also affects the online visualization. You can only display characteristics that are processed by the logic of the PDV data collection and are not only stored.

Formula parameters only

Use this option to define that the created process parameter is only used as parameter to calculate a further process parameter. If you set this option for a process parameter, this process parameter will never assess measured values with respect to violated limit values and these process parameters will never be stored in the database.

Compute limit values

If you enable this option, the fields for the calculation of target values or limits are activated. Formula calculation, see section "Formula calculation"

Upper TL formula

You can use the formula for the calculation of the UTL to calculate a new UTL in combination with other parameters. If you recalculate the UTL, this will also lead to changed target values. See section 3.5 for further information on formulas.

Upper PAL formula

You can use the formula for the calculation of the UPAL to calculate a new UPAL in combination with other parameters. If you recalculate the UPAL, this will also lead to a changed target value. See section 3.5 for further information on formulas.

Target value formula

You can use the formula for the calculation of the target value to calculate a new target value in combination with other parameters. If you recalculate the target value, this will also lead to a changed target value. See section 3.5 for further information on formulas.

Lower PAL formula

You can use the formula for the calculation of the LPAL to calculate a new LPAL in combination with other parameters. If you recalculate the LPAL, this will also lead to a changed target value. See section 3.5 for further information on formulas.

Lower TL formula

You can use the formula for the calculation of the LTL to calculate a new LTL in combination with other parameters. If you recalculate the LTL, this will also lead to a changed target value. See section 3.5 for further information on formulas.

Formula calculation

Using the formula calculation function, you can calculate the measured values collected via machine connection before you evaluate them. For example, you can add an offset to the calculation of a characteristic, you can directly convert units in the collection process or you can perform calculations in combination with other characteristics.

To store and display a formula, the user must have the function authorization iriscp.formula. If the user is not authorized, the field is not displayed.

You specify the calculation of measured values in the application *Characteristics* in tab *General*. You can also calculate the target values of an automatically collected characteristic and the upper/lower tolerance or action limits.

Important: HYDRA versions below MES Weaver 2.0 with server systems AIX, HP-UNIX, SCO-UNIX or DEC ALPHA do not provide this kind of formula calculation.

The first part of the formula specifies the level where the formula calculation is made. The following types are available:

- V - Calculation on the level of single values without self-reference.
  For each measured value of the characteristics involved, exactly one single value is generated for the calculated characteristic.

- O - Calculation on the level of samples with self-reference.
  Use this type of formula to refer to the value itself and to integrate limit or target value.

After the above identifier, the actual formula is specified. The following operators, functions and constants are supported:

Functions	
abs(x)	Calculates the absolute value
atan(x)	Calculates the arc tangent
cosh(x)	Calculates the hyperbolic cosine
float(x)	Converts the value into a floating point number
sqrt(x)	Calculates the square root
acos(x)	Calculates the arc cosine
atan2(y,x)	Calculates the arc tangent of y/x
exp(x)	Calculates the exponential value
log(x)	Calculates the natural logarithm
sin(x)	Calculates the sine
tan(x)	Calculates the tangent
asin(x)	Calculates the arc sine
cos(x)	Calculates the cosine
int(x)	Converts the value into an integer
log10(x)	Calculates the common logarithm
round(x)	Rounds to integer value
round(x,y)	Rounds the value x to y decimal places
sinh(x)	Calculates the hyperbolic sine
tanh(x)	Calculates the hyperbolic tangent
trunc(x)	Reduces the value x to an integer value
trunc(x,y)	Reduces the value x to y decimal places
Operators	
x + y	Addition
x – y	Subtraction
x / y	Division
x * y	Multiplication
x ** y	Calculates x to the power of y
Constants	
pi	`3.141592654`
e	`2.718281828`

If constant numeric values are used in formulas, you must be careful not to use thousand separators. If these constants are floating point numbers, be careful to use a dot as decimal separator instead of a comma.

The following syntax `[A:B:C]`. applies for the variables that identify the single or default values of the process parameters involved.The available values are listed below.

You can specify the following values for section `A`:

- `X` – single value/measured value
- `UTL` – Upper Tolerance Limit
- `UPL` – Upper Process Action Limit
- `TV` – Target Value
- `LPL`– Lower Process Action Limit
- `LTL` – Lower Tolerance Limit

Note: The value X defined for section A (single value/measured value) is only used for the calculation of measured values; this means, the value is not used in the target value formulas or formulas of the upper/lower tolerance or action limits.

Note: Aggregate functions like MAX, MIN or AVG, which are used in HYDRA CAQ, are not supported with automatically collected characteristics.

Section `B` describes how the relevant characteristic is identified. The following possibilities are available:

- `SELF – self-reference`
  `(This requires formula type O)`
- `PPARAM – Reference to further process parameters of the same machine`
  `With this formula, only process parameters of the same machine can be calculated.`

Section C identifies the characteristic using the field content specified in section B. This means: If a self-reference is specified in section B, a section C may not exist. If instead a reference to a further process parameter is stored for PPARAM, the process parameter must be specified in section C.

Example 1:

The process parameter always collects ten times the rounded value from the machine connection.

→ Formula: `O: round([X:SELF]) * 10`

Example 2:

The characteristic "area" results from the product of the process parameters `LENGTH (LAENGE)` and `WIDTH (BREITE)`. For each single value of the two source characteristics, a single value is calculated for the characteristic "area".

→ Formula: `V: [X:PPARAM:LAENGE] * [X:PPARAM:BREITE]`

Example 3:

The following formula is not stored for the measured value, but for the target value of the process parameter "speed". The target value is calculated using the own target value plus the target value of process parameter DURCHMESSER (diameter). The target value for "speed" is therefore recalculated, if the target value for "diameter" changes.

→ Formula: `O: [TV:SELF] + [TV:PPARAM:DURCHMESSER]`

Note:

The formula calculation is performed directly after reception of the measured values from the PDV data collection. Changes to any of the process parameters included in the formula are then used as events that trigger recalculation.

4 PDV Events

Summary

Menu	Master Data → Process Data Processing → Events
Transaction code	peve
Function authorization	peve

Utilization

With PDV events, various events that can occur on the machine can be configured and collected. In addition, the configured PDV events can be identified as an alarm, which can create an appropriate output signal if it occurs on the machine.

The assignment of the PDV events to a physical channel on a machine is performed by configuring the logical channels. A log, or the time-related documentation of the events that occur, can be carried out using the protocol dialog described in the "Process events" application.

PLEASE NOTE!
To do this, in the configuration of the logical channels both the type (`E`) and the related control must be correctly parameterized.

Selection parameters

The following selection criteria are available in the respective application:

Event ID

Option to search for an event identification with a wildcard function.

Designation

Option to search for an event designation with a wildcard function.

Field description

The PDV event configuration contains the following information:

Event ID

Unique identification of the configured event

Designation

Optional parameter with which a designation can be assigned to an event configuration

Alert

Identifier as to whether or not the event is to trigger an alarm (physical signal) when it occurs

Alert duration

Delay time specifying how long after the event has occurred, the alert is to remain at the alarm channel. The unit is entered in seconds.

Event type

Declaration of the event type: entered as event (F) or as hint (H). If no type is defined, the default type (P) will be entered.

Event category

Category to classify events (also referred to as malfunctions) and notes e.g. in the evaluation of process events.

5 ID Tracing (tabular)

Overview

Menu	Quality management → Process analysis → ID tracing
Transaction code	cidt
Function authorization	cidt

This document describes the application "ID Tracing (tabular)" in the Manufacturing Operation Center (MOC).

Usage

ID Tracing enables the tabular presentation and analysis of process values that can be selected referring to search keys (IDs). Search keys are identification tags provided by the machine. They are used to identify measurement tuples instead of or in addition to the machine and time stamps of data collected in the database.

Integration

Measured values must be collected and saved based on IDs in order to use this function. For this purpose, at least one channel with the data type "tag" must be defined in the application "PDV - logical channels" and data collection must be configured accordingly.

Selection parameters

IDs can be selected in the selection panel. The following selection criteria are available in the application:

Tag type:

Identifies the key field. The name of the ID tag.

Tag ID

Search value of the selected ID tag.

Workplace

Number of the workplace as an additional search field. Is required, if the same tags are available at different machines/workplaces.

Time range from - to:

The data selected by the tag value is restricted temporally.

Consider the last time range of the tag value only

If this function is enabled the data selected is restricted to last time range recorded for the selected tag value.

Field descriptions

This tabular report shows the process characteristics recorded and saved in the database including the following information:

Characteristic

Technical name of the characteristic or the entered process parameter

Designation

Defined characteristic name

Machine

Defined machine

Process parameters

Defined process parameter

Target value

The target value defined for this recorded process parameter at that specific point in time

UTL

The upper tolerance limit defined for this recorded process parameter at that specific point in time

UPAL

The upper process action limit defined for this recorded process parameter at that specific point in time

LTL

The lower tolerance limit defined for this recorded process parameter at that specific point in time

LPAL

The lower process action limit defined for this recorded process parameter at that specific point in time

Unit

Unit of the recorded characteristic

Measured value

Measured value recorded for the characteristic at the time of measurement

Time of measurement

Time of measurement of the characteristic